AI Ethics

Course Num: SEIS 651

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Week 5

Agenda

Focus Area

- AI + Ethics: Neural Networks

This is my favorite week because…

Intersection of technical and philosophical thinking.

So much of this is misunderstood … which, ironically, is perfect for our class!

Week 5

Outline – Key Points

- Overview of neural networks and the black box problem
- Explainable vs. Understandable AI
- Ethical considerations
- Risk mitigation
- Exercise: Applying NIST AI RMF

Lecture

Recall…

"To study choice and purpose is psychology; to study choice as affected by the rights of others and to judge it as right or wrong by this standard is ethics."

Dewey (1908)

Consider…

Can we effectively judge a “choice” made by an AI model as “right” or “wrong” if we don’t fully understand that model?

Neural Networks

- Intersection of AI and Ethics: Neural Networks
- Neural Networks are inspired by what we understand about how human neurons work
- There are SEVERAL types of neural network models, not just one

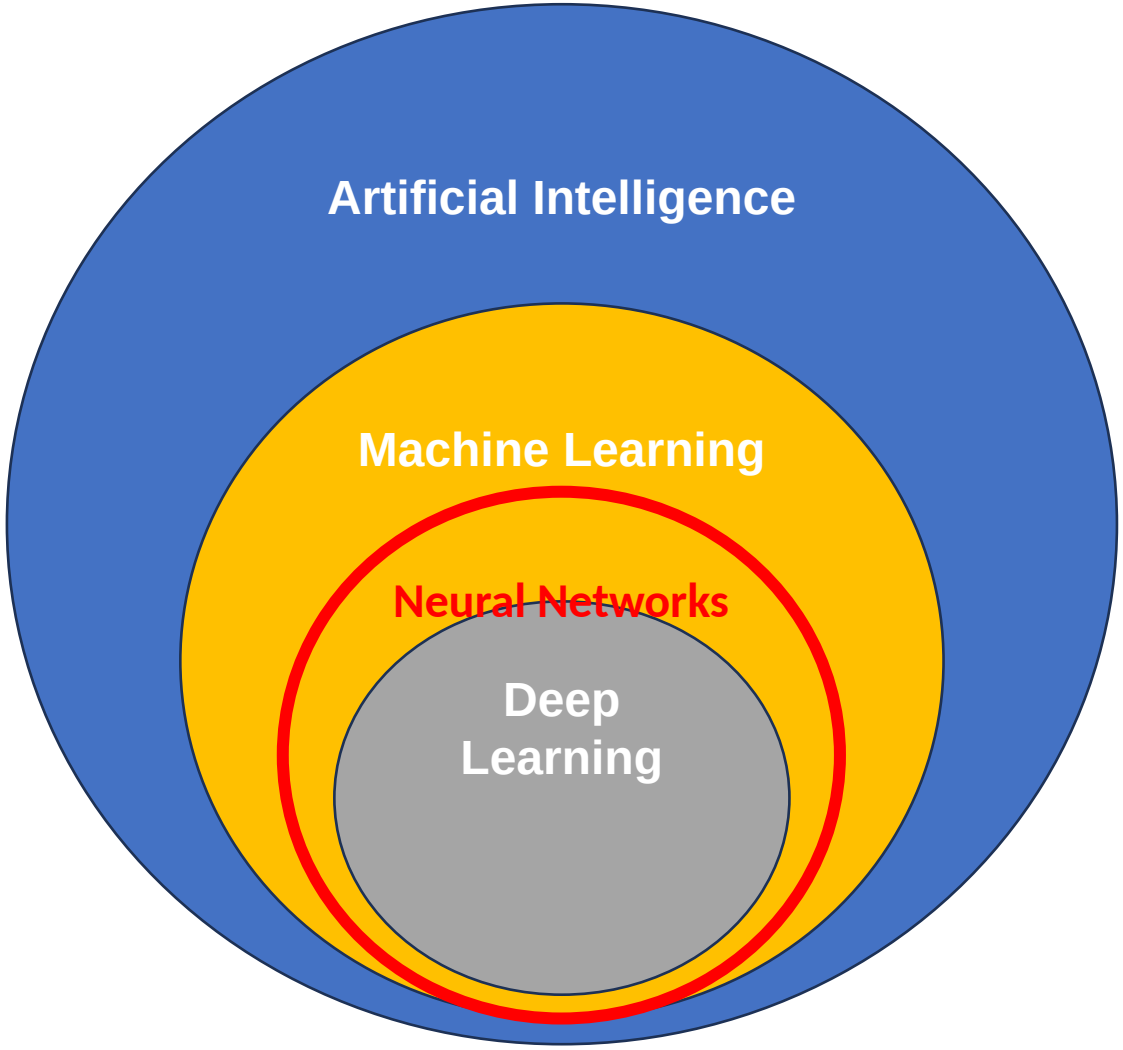


Image generated with Dall-E using the prompt: "A hyper realistic drawing of a neural network represented by an iceberg with its nodes below the waterline."

Recall our AI taxonomy...

- Neural networks are machine learning algorithms, AI solutions

- Deep learning is achieved through neural networks (but not all neural networks are deep learning)



Recall…





















STITCH FIX

VERY BASIC Structure

- A defining characteristic of a neural network is its "hidden" layers
  - Input >> Hidden layer(s) >> Output
    - Deep learning involves several hidden layers
- These layers are made up of nodes (“computational unit”) that are weighted and connected
  - Connections, weights, hidden nodes, etc., are defined algorithmically

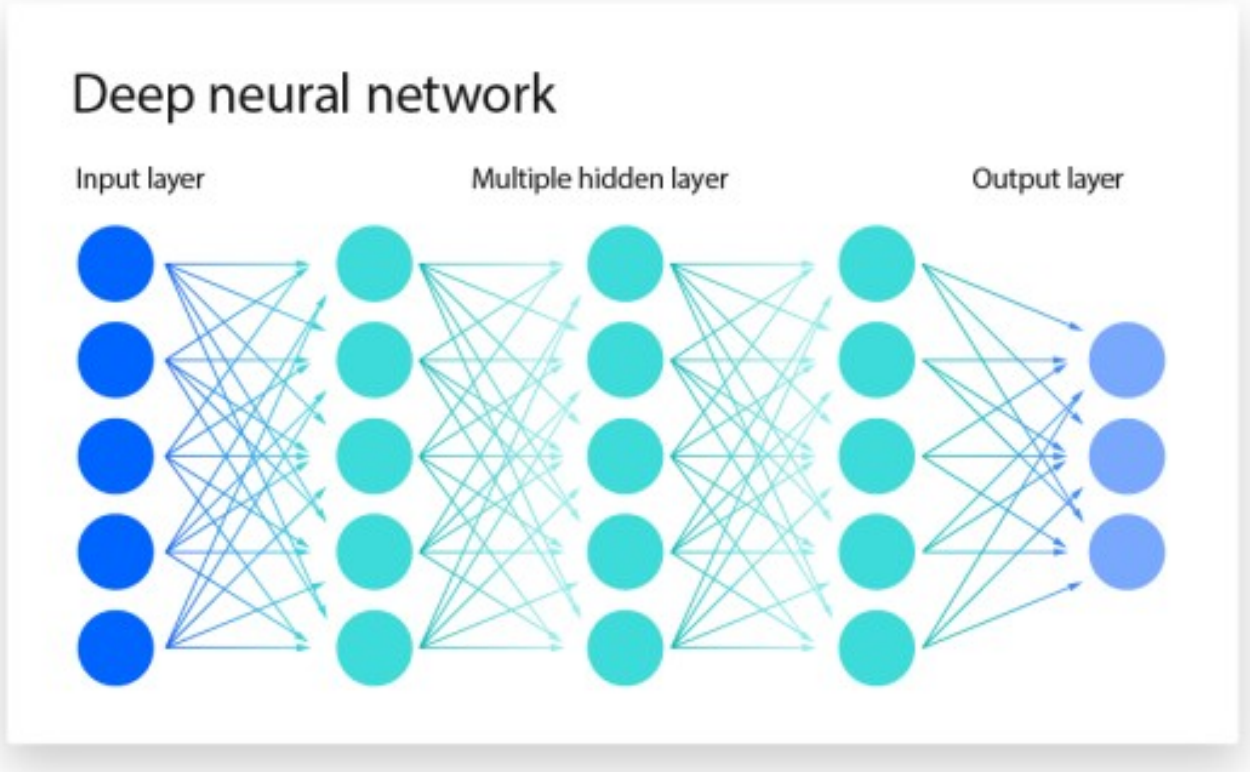


Image source: https://www.ibm.com/topics/neural-networks

More BASIC Structure

- Nodes don't have to line up with the input layer
- Weights are not necessarily linear
- Network can be traversed in both the forward and backward direction, in cycles

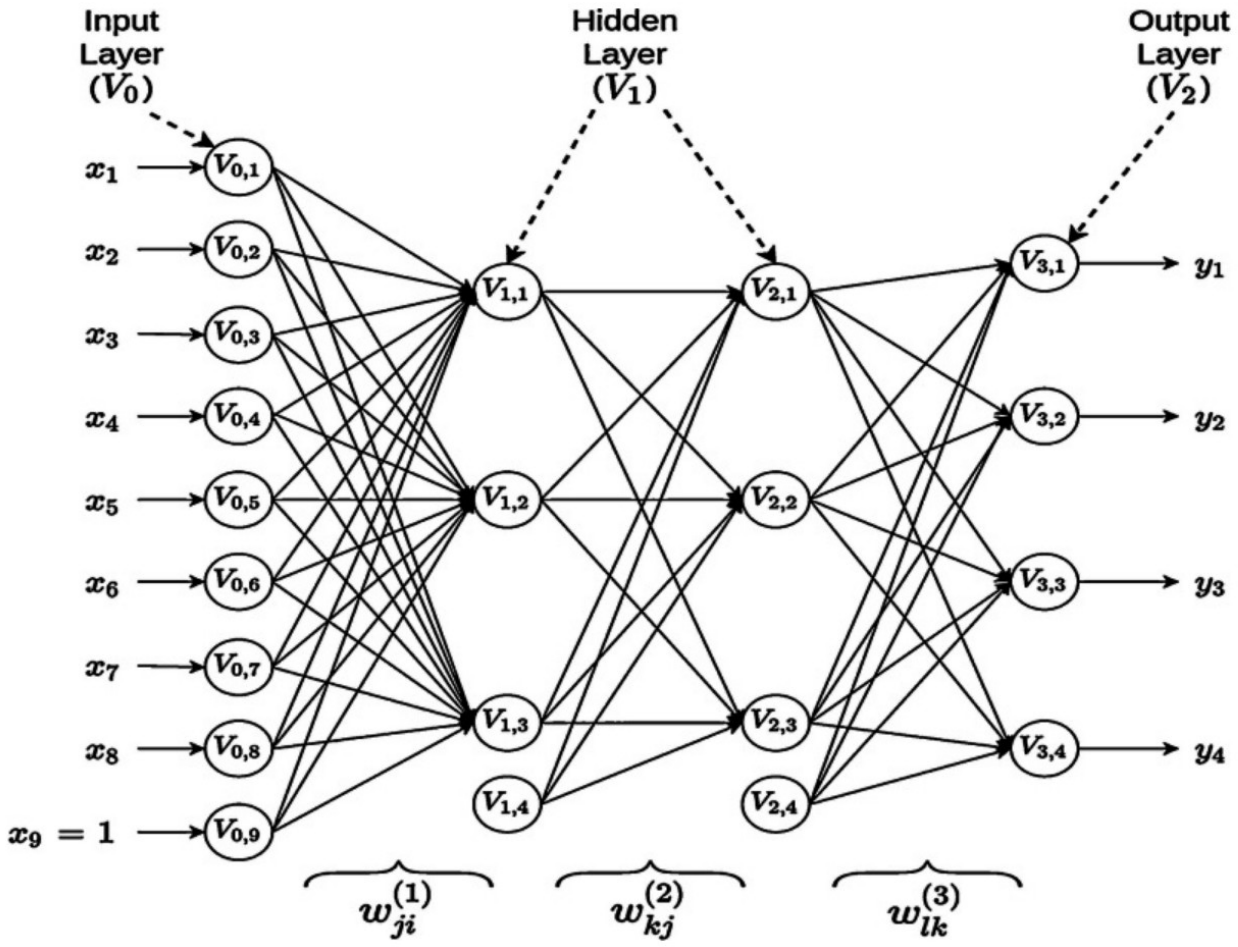


Image Source: https://www.ncbi.nlm.nih.gov/books/NBK583971/ (fig 10.4)

What are they good at?

- Analyzing complex, large, and/or unstructured data (3Vs!). Examples:
  - Images, Audio, Video
  - Language
  - Graphs

- Great at making predictions from datasets that other models just can’t handle. Examples:
  - Generative AI
  - Directions and routing
  - Image classification
  - Voice and facial recognition

- **Key Takeaway**: Neural networks can be extremely powerful!

Other Considerations

- Neural networks are not necessary for all AI/ML/Data Science problems
  - e.g., Making predictions based on small training datasets

- Resource intensive
  - *They need a LOT of data and can take FOREVER to train*
  - May come with energy usage concerns
    - e.g., https://www.osti.gov/servlets/purl/3015048 (2026)

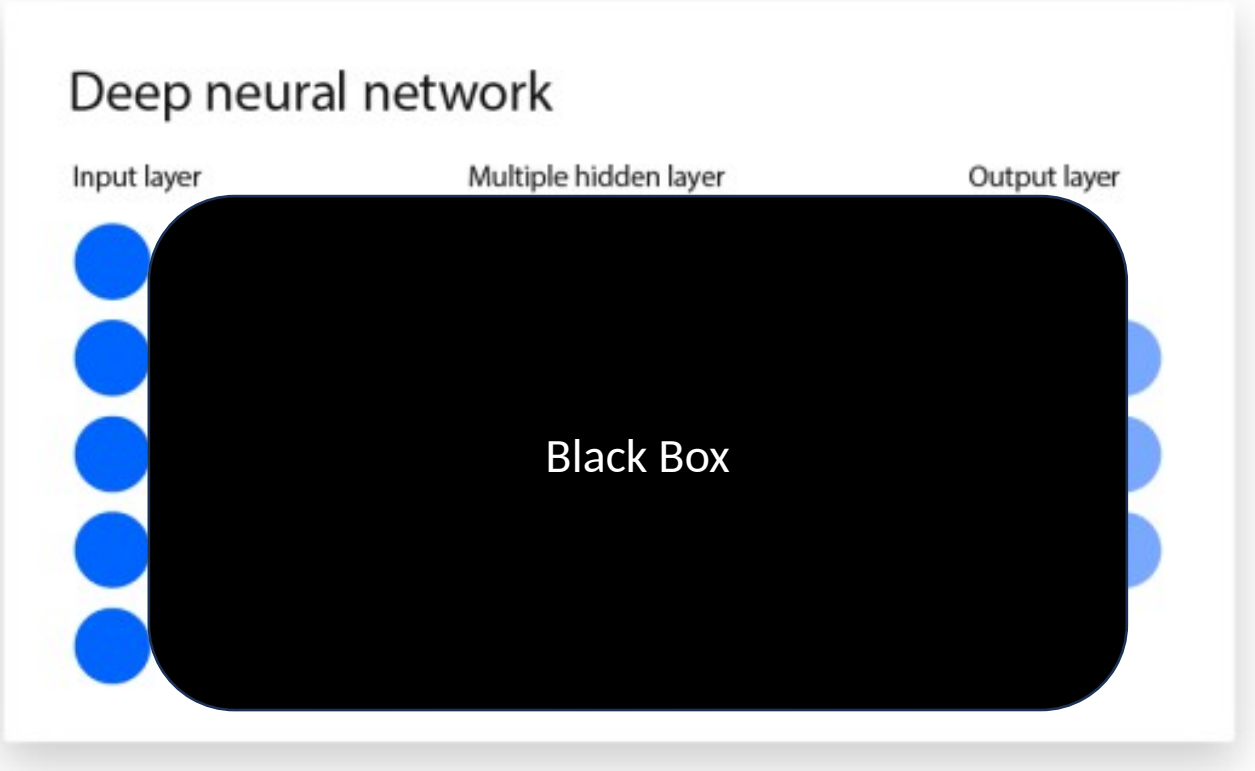
- Vulnerable to data poisoning

- **Key takeaways**: Choose the right model for your problem and beware of overengineering!

The Black Box Problem

- Additionally, neural network models are not transparent – as if they are functioning inside of an opaque "**black box**."

- This term illustrates how neural network models can be extremely difficult (and sometimes impossible!) to interpret and explain. Meaning we don't know *exactly* what a NN is doing!

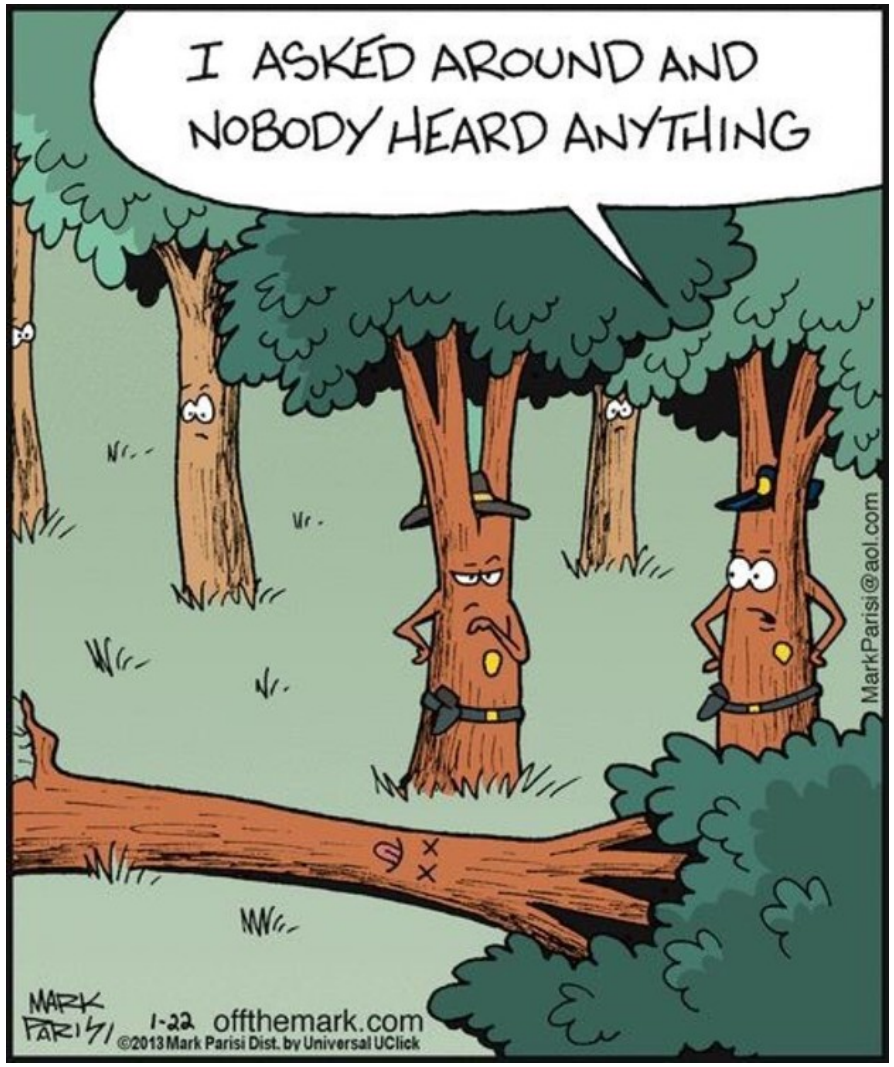


Explainability

- Explainable AI is one response to the black box problem
  - Implementing a “glass box” rather than a “black box”
  - Using explainable AI methods (SHAP, LIME, a bunch more)

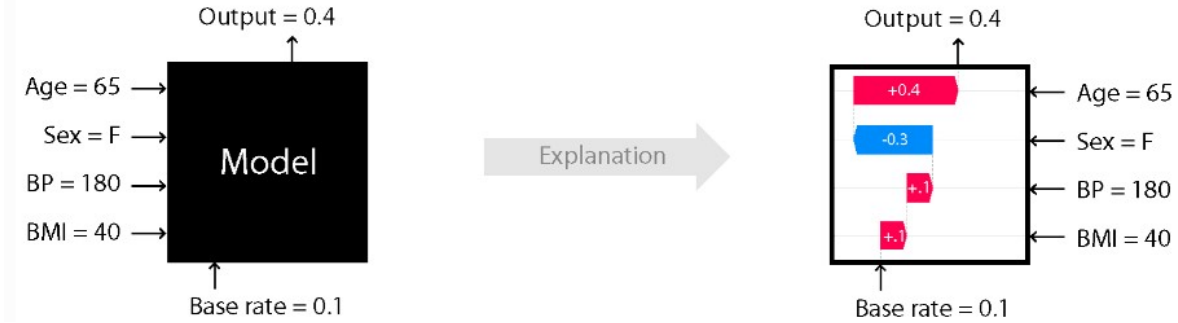
- But not all models are “explainable”
  - Developers and engineers don’t always know what the model is explicitly doing.
  - Some models are simply too large or too complex for our brains to conceptualize!
    - Deep neural networks can have millions or even billions of nodes and weights!

- Even if you can perfectly explain your model, can your user base understand it? Does this matter?

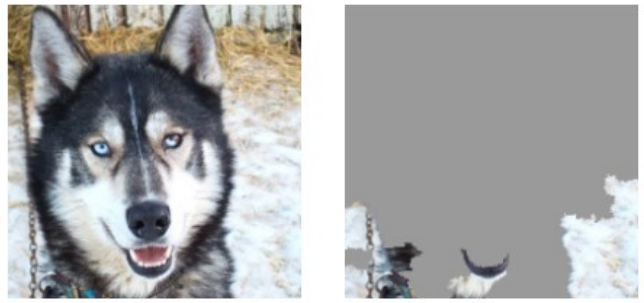


Explainable AI Examples

- There are methods that can be used to help explain what a complex model is doing
  - E.g., SHAP
  - E.g., LIME
- These may not be explaining exactly what each neuron is doing, but it is giving an explanation in terms of the important inputs which may be more helpful to consumers, stakeholders, etc.



SHAP Image taken from: https://shap.readthedocs.io/en/latest/



(a) Husky classified as wolf (b) Explanation

Figure 11: Raw data and explanation of a bad model's prediction in the "Husky vs Wolf" task.

LIME Image taken from: https://arxiv.org/pdf/1602.04938

Explainable AI Limitations

- Example: SHAP
  - “To address these potential dependencies, the Shapley value requires that the outcome of each possible combination (or coalition) of features must be considered to determine the importance of each feature. Given *d* features, there are 2^d such possible feature combinations, each corresponding to a potential model.” …“However, even for reasonable values of *d*, say 50 features, it is computationally prohibitive and impractical to train 2^d possible models. As a result, SageMaker Clarify needs to make use of various approximation techniques.” AWS Documentation

- 2^50 = 1,125,899,910,000,000 models!
- Does this mean SHAP is not a viable solution for larger models? (No! … So what *does* this mean?)

(Some) Ethical Implications

Do we always ***need*** to explain these models? In short, no. But consider…

- Informed consent
  - Can a person provide informed consent for their data to be processed in a model they can't understand?
- Compliance
  - Can we ensure that a model complies with laws/regulations/standards, etc., if we don't completely know what it's doing? Are we required by law/regulation/policy to justify our actions?
- Harm reduction
  - Can we be confident that a model's output will not cause harm if we're not entirely sure what the model will produce?
- Liability and Risk
  - Who is responsible if something goes wrong? Should a developer or company be held liable if they didn't explicitly program the model to behave a certain way? How likely is a negative outcome/impact?

Perspective

- **AI/ML developers, engineers, practitioners:**
  - The need for explainability and transparency may influence your choice of model/algorithm, and how you monitor/maintain it
- **Consumers:**
  - Are you comfortable not knowing how your data is being used, or how an AI-driven decision was made?
  - *Who gets to decide what consumers should be comfortable with?*
- **Companies:**
  - Companies may need to weigh their appetite and tolerance of risk, implement mitigation strategies, etc., before committing to a black box solution
- **Policy (and Law) Makers:**
  - Does the use of these models need to be regulated? Does your organization have a governance strategy in place?

Risk Mitigation

- “In the context of the AI RMF, risk refers to the composite measure of an event’s **probability** of occurring and the **magnitude** or degree of the consequences of the corresponding event.” (*NIST AI RMF 1.0, emphasis added*)
  - Note, events can be “positive, negative, or both”
  - The AI RMF “… offers approaches to **minimize anticipated negative impacts** of AI systems and identify opportunities to **maximize positive impacts**.”
- In practice, you will need to decide **what risks are acceptable.**

Back to Dewey…

Can we effectively judge a “choice” made by an AI model as “right” or “wrong” if we don’t fully understand that model?

Recap – Week 5

- Neural networks are extremely powerful, and excel in areas that other models can’t address

- Neural networks are an example of “black box” AI solutions that are difficult (or impossible) to understand and explain (even to the developers!)

- This presents both practical and ethical challenges.
  - Note: This does not mean that neural networks or other black box models are inherently unethical!
  - What risks are acceptable?

References

- https://www.ncbi.nlm.nih.gov/books/NBK583971/
- AWS SageMaker Clarify - Shapely Values
- SHAP, LIME, CAM - Explainable AI
- LIME Paper
- AWS RNN Explanation
- Language Models are Few-Shot Learners (Open AI paper, 2020)

Optional Further Reading

- https://www.bbc.com/future/article/20230405-why-ai-is-becoming-impossible-for-humans-to-understand
- https://www.ncbi.nlm.nih.gov/books/NBK602381/

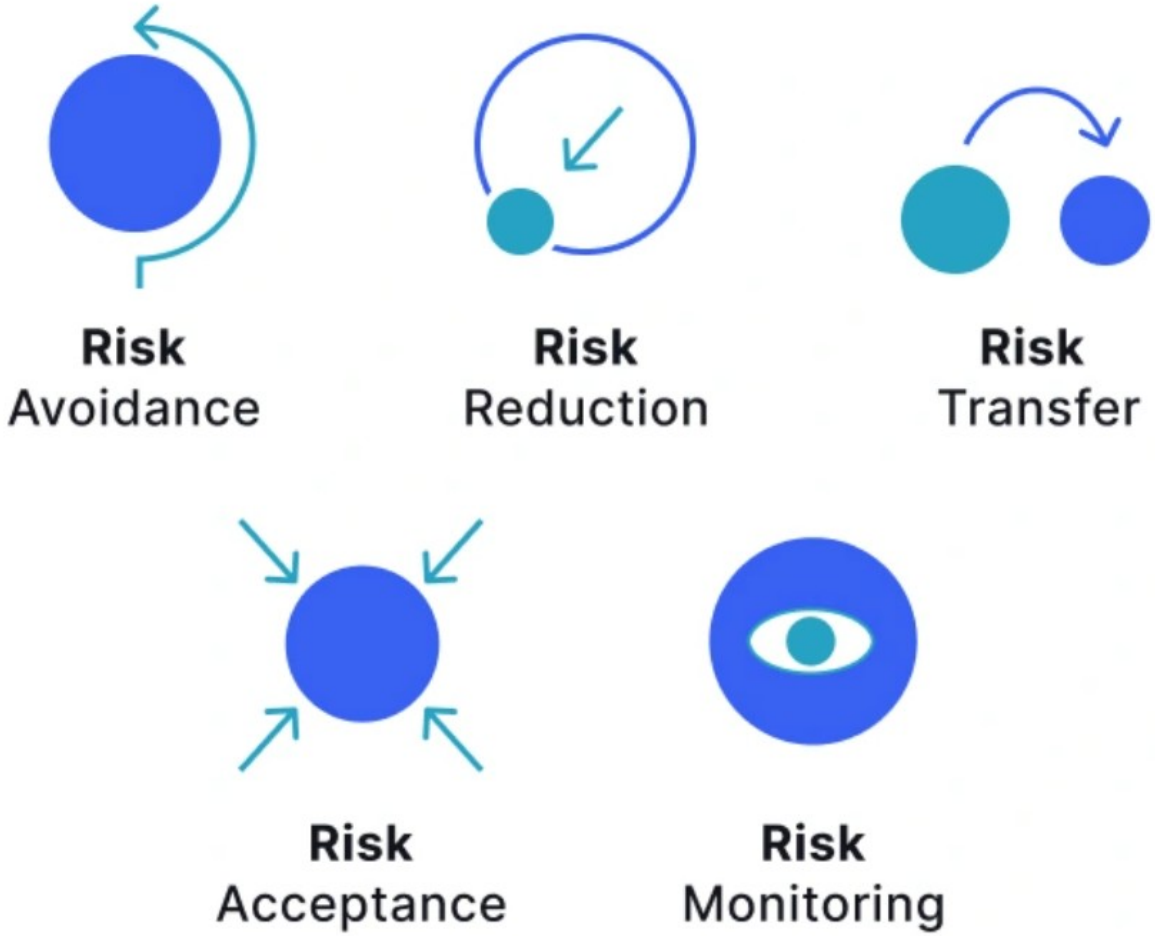
Exercise

Applying NIST AI RMF to a black box AI system

Risk Mitigation

- Risk Mitigation doesn't mean avoiding every risk every time

- Mitigating risk might look more like *expecting* the event to happen, and planning for it
  - Also planning for "unknown unknowns"



https://www.everbridge.com/blog/risk-mitigation/

Black Box Risk Mitigation



- In addition to the above risk mitigation strategies…
- Choose the best model for your use case
  - Consider the need for explainability as part of your use case
    - Evaluate in context of your customer/user base, stakeholders, legal counsel, etc.
    - Beware of overengineered solutions ☺
- Stakeholder engagement and education
- Monitor the model
- Consider explainable AI solutions

Exercise

(Totally Hypothetical!) Scenario:

Imagine you are members of the University's AI Governance Board. The University is considering providing an image recognition/interpretation tool called **Tutor651** to all students across all programs. The tool allows students to upload an image of their written notes, and produces flashcards and other study materials based on those images. Your job is to evaluate the risks associated with **Tutor651** and propose mitigation strategies that the University can implement.

Task: https://airc.nist.gov/airmf-resources/airmf/5-sec-core/

In your group, review your assigned AI RMF Core Function using the link above. Choose one category or subcategory from your function and **propose one risk mitigation practice** that the University could implement based on this function. **(Be prepared to share your group's proposed practice with the class.)**

Exercise

Group Assignments

- **Group 1 - Govern:** Cross-functional, creates a "culture of risk management."
- **Group 2 - Map:** Establish context. Incorporates knowledge of the AI system and the organization's goals.
- **Group 3 - Measure:** Analyze risk, assess impact across stakeholder groups. Establishes metrics, testing procedures, feedback mechanisms.
- **Group 4 - Manage:** Allocation and coordination of resources, incident response planning, communication.

